

In-class assignment 4 (Individual): Introduction to Jenkins pipeline

AD/ SPRINT 3 / 2

Fall 2026

Note: The in-class assignments are compulsory for completing both the sprint implementation and the overall project. Failure to complete the in-class assignments will result in losing the implementation part of your sprint tasks and, consequently, your overall course grade. **This will cause you to fail the course if your total points do not reach the required threshold of 2 points.**

This assignment required that you participate lecture and do the class exercise at the same time with teacher

Tasks Objective

Based on your Week 5 in-class assignment (Temperature Converter), implement the following steps and submit the results in Oma:

1. Create a Jenkinsfile (pipeline) to generate a JaCoCo report.
2. Build and run the local Docker image of your application and verify that everything is working correctly.
3. Extend the pipeline to deploy the image to *hub.docker.com*.

Instructions

Step 1. Add the jenkinsfile into your Maven project

right click in the project (inteliJ)---> new --- >file

Name: Jenkinsfile

write the stages in the pipeline in your Jenkinsfile

Note: You may copy and paste the following code but make sure you fix before commit

```
pipeline {
  agent any
  stages {
    stage('Checkout') {
      steps {
        git 'https://github.com/your-username/your-repo.git'
      }
    }
    stage('Build') {
      steps {
        bat 'mvn clean install' // sh for linux and ios
      }
    }
    stage('Test') {
      steps {
        bat 'mvn test'
      }
    }
    stage('Code Coverage') {
      steps {
        bat 'mvn jacoco:report'
      }
    }
    stage('Publish Test Results') {
      steps {
        junit '**/target/surefire-reports/*.xml'
```

```

    }
  }
  stage('Publish Coverage Report') {
    steps {
      jacoco()
    }
  }
  // follow the lecture demo for hub.docker.com deployment stages
}
}

```

Step 2: Commit the final version of your source code after modifying the pipeline

Step 3. Create a local Jenkins project

- *Select New Item.*
- *Select pipeline project and give it a name (e.g., Amir_Temperature_V1_pipeline) then ok.*
- *Select pipeline, in the definition ---> select pipeline Script from SCM. --> select git - Add the GitHub repository URL.*
 - *under the Script Path add---> Jenkinsfile*
 - *Save*
- *Run the project through Build now*

Submission Instructions

1. GitHub repository link
2. Screenshot of a successfully executed pipeline (e.g., Ocean Blue view or stage view). See the sample output from class demo (SVG_301x.java)
3. Screenshot of the generated image in your hub.docker.com
4. Screenshots showing the successful execution of the image